List Colouring the Odd Torus

A step-by-step account of what is proved, what is computed, and what is still open

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Version of 28 October 2026, 2 PM

Every computer-assisted claim names the script in `~/work/gt/odd-torus/` that reproduces it (Appendix A).

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How to read this book

Results are numbered by chapter (Lemma 4.5 is the fifth numbered item of Chapter 4) and are reused by number later. The colour of a box tells you how the result is established:

Known result	published by others; we cite it and do not prove it
Theorem / Lemma	proved in this book by hand
Computer-assisted result	proved, but one step is an exact computation (script named)
Verified, not proved	true in every case we tested; no proof yet
Open	not known

To check our work, read a chapter, then try its "Think it over" questions. Chapters 1 and 2 are background; Chapter 3 is the main finished theorem; Chapter 4 has lemmas for every odd m; Chapters 5 and 6 are about the open case $m = 5$; Chapter 7 records dead ends.

Notation used throughout.

C_k	the cycle with vertices $0, 1, \ldots, k-1$
$C_m \,\square\, C_n$	the torus grid: vertex (i, j), row $i \in \mathbb{Z}_m$, column $j \in \mathbb{Z}_n$
$L(v)$	the list of allowed colours at v; always $\lvert L(v)\rvert = 3$ unless stated
palette	the set of all colours appearing in any list
ω	$e^{2\pi i/3}$, a cube root of unity

Chapter 1

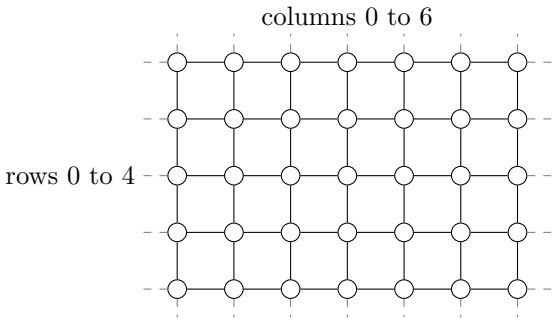
The problem

1.1 List colouring

Definition 1.1 (List colouring). Let G be a graph and let every vertex v have a list $L(v)$ of colours. An *L-colouring* is a choice of one colour $c(v) \in L(v)$ for each vertex, such that adjacent vertices get different colours. G is *k-choosable* if it has an L-colouring for *every* choice of lists with $|L(v)| = k$. The smallest such k is the *choice number* $\chi_\ell(G)$.

Ordinary colouring is the special case where all lists are equal. So $\chi_\ell(G) \geq \chi(G)$, and the difference can be large: the adversary chooses the lists.

Definition 1.2 (Torus grid). For $m, n \geq 3$, $C_m \,\square\, C_n$ has vertices (i, j) with $i \in \mathbb{Z}_m$ (the *row*) and $j \in \mathbb{Z}_n$ (the *column*). The vertex (i, j) is adjacent to $(i \pm 1, j)$ (its *column neighbours*) and to $(i, j \pm 1)$ (its *row neighbours*). Every vertex has degree 4. Each column is a copy of C_m; each row is a copy of C_n.



$C_5 \,\square\, C_7$. Dashed edges wrap around.

1.2 The conjecture and what is known

Known result 1.3 (Cai, Wang and Zhu [CWZ])

Conjecture. Every torus grid $C_m \,\square\, C_n$ is 3-choosable.

Known result 1.4 (Li, Shao, Petrov and Gordeev [LSPG])

If m or n is even, $C_m \,\square\, C_n$ is 3-choosable. If m and n are both odd, the Alon–Tarsi number of $C_m \,\square\, C_n$ is 4.

So the open case is m, n both odd. The second sentence matters: the standard algebraic method (Section 2.2) *cannot* prove 3-choosability for odd tori. Any proof must contain a step that looks at the actual lists.

Known result 1.5 (Galvin [Gal])

$C_3 \,\square\, C_3$ is 3-choosable. (It is the line graph of $K_{3,3}$, and Galvin proved that line graphs of bipartite graphs have choice number equal to their chromatic number.)

Known result 1.6 (Sabidussi [Sab])

$\chi(G \,\square\, H) = \max(\chi(G), \chi(H))$. In particular every odd torus $C_m \,\square\, C_n$ has a proper colouring with 3 colours.

Think it over

Q1. Show directly that $\chi_\ell(C_3 \,\square\, C_n) \geq 3$. (Hint: columns.)

Q2. Lists of size 5 are always enough for a 4-regular graph. Why? (Colour greedily.)

Q3. Why does $C_m \,\square\, C_n$ have exactly twice as many edges as vertices?

Chapter 2

Tools from the literature

This chapter lists every published result we use. Nothing here is ours.

2.1 Cycles and degree lists

Known result 2.1 (Erdős, Rubin and Taylor [ERT]; Borodin [Bor])

Let G be connected and $|L(v)| \geq \deg(v)$ for every vertex. Then G is L-colourable, unless every block of G is a complete graph or an odd cycle (a "Gallai tree").

Two special cases are used again and again.

Known result 2.2 (cycles; special case of 2.1)

(a) A cycle whose lists all have size at least 2 can be coloured, unless the cycle is odd and all its lists are the same 2-set.
(b) A cycle with one list of size 3 and the others of size at least 2 can always be coloured.

2.2 The polynomial method

For a graph G with a fixed orientation of its edges, the *graph polynomial* is $P_G = \prod_{(u,v)}(x_u - x_v)$, one factor per arc. A colouring is proper exactly when $P_G \neq 0$.

Known result 2.3 (Combinatorial Nullstellensatz, Alon [Alon])

Let $\deg P = \sum_v t_v$ and suppose the coefficient of $\prod_v x_v^{t_v}$ in P is nonzero. If $|S_v| > t_v$ for every v, then P is nonzero somewhere on $\prod_v S_v$.

Applied to P_G: if that coefficient is nonzero, G can be coloured from any lists with $|L(v)| \geq t_v + 1$.

Known result 2.4 (Coefficient formula, Lasoń [Lason]; Karasev and Petrov [KP])

If $\deg P = \sum t_v$ and $|A_v| = t_v + 1$, then $[\prod x_v^{t_v}]\, P = \sum_{a \in \prod A_v} P(a) / \prod_v \varphi_v'(a_v)$, where $\varphi_v(z) = \prod_{b \in A_v}(z - b)$.

Known result 2.5 (Alon and Tarsi [AT])

Let D be an orientation of G. If the numbers of even and odd Eulerian subgraphs of D differ, then G is L-colourable whenever $|L(v)| > \mathrm{outdeg}_D(v)$ for every v.

Note. In a bipartite graph every cycle is even, so every Eulerian subgraph has an even number of edges. Hence there are no odd Eulerian subgraphs, while the empty subgraph is even: *every* orientation of a bipartite graph satisfies the hypothesis of 2.5.

Known result 2.6 (Hakimi [Hak])

A graph has an orientation with $\mathrm{outdeg}(v) \le k(v)$ for all v if and only if every subgraph H satisfies $|E(H)| \le \sum_{v \in H} k(v)$.

Think it over

Q1. Prove 2.2(a) yourself: colour around the cycle and see where you get stuck.

Q2. Why is the condition in 2.6 obviously necessary?

Q3. Using 2.5 and the Note, show that an even cycle is 2-choosable.

Chapter 3

The finished case: $C_3 \square C_n$

3.1 Spare room and shapes

Every cut in this book uses 2.3 in the same way. This section states that use precisely and proves it. Fix an orientation of a graph H and write $P_H = \prod_{(u,v)}(x_u - x_v)$ for its graph polynomial.

Lemma 3.1

P_H is a nonzero homogeneous polynomial of degree $|E(H)|$.

Proof. Each factor $x_u - x_v$ is nonzero and homogeneous of degree 1. A product of nonzero polynomials is nonzero, and a product of homogeneous polynomials is homogeneous of degree equal to the sum of the degrees. There is one factor per edge. □

Definition 3.2 (spare room, shape). Let H have N vertices. Its *spare room* is $d(H) = 2N - |E(H)|$. A *shape* for H is a map $\mu : V(H) \to \{0, 1, 2\}$ with $\sum_v \mu(v) = d(H)$; we say μ *spends* $\mu(v)$ units at v. The *coefficient* of the shape is $c_\mu = \left[\prod_v x_v^{2-\mu(v)}\right] P_H$.

The units are not a gift: if $d(H) > 0$, the monomial $\prod_v x_v^2$ has degree $2N > \deg P_H$, so its coefficient is 0 and exponents must be lowered by exactly $d(H)$ in total. The choice is *where*.

Lemma 3.3 (shape lemma)

Let μ be a shape for H with $c_\mu \neq 0$. Let each vertex have a list $L(v)$ of 3 colours, and choose any subsets $L'(v) \subseteq L(v)$ with $|L'(v)| = 3 - \mu(v)$. Then H has a proper colouring c with $c(v) \in L'(v)$ for every v.

Proof. **Step 1.** Put $t_v = 2 - \mu(v) \geq 0$. Then $\sum_v t_v = 2N - d(H) = |E(H)| = \deg P_H$ by 3.1.

Step 2. Map the colours injectively into $\mathbb{C}$; then $L'(v)$ becomes a set $S_v \subseteq \mathbb{C}$ with $|S_v| = 3 - \mu(v) = t_v + 1 > t_v$.

Step 3. The coefficient of $\prod_v x_v^{t_v}$ is $c_\mu \neq 0$, so by 2.3 some $s \in \prod_v S_v$ has $P_H(s) \neq 0$.

Step 4. $P_H(s) \neq 0$ means $s_u \neq s_v$ on every arc, so $c(v) = s_v$ is a proper colouring with $c(v) \in L'(v)$. □

Note. This is what spending means. One unit at v ($\mu(v) = 1$): choose $L'(v) = L(v) \setminus \{\beta\}$, so the colouring **forbids** β at v. Two units ($\mu(v) = 2$): choose $L'(v) = \{\alpha\}$, so the colouring **pins**

v to α. The lemma promises *some* colouring obeying these choices and nothing more; the rest of every cut argument must therefore work against *every* such colouring.

Corollary 3.4 (adjacent pins)

If $\mu(u) = \mu(v) = 2$ for adjacent u, v, then $c_\mu = 0$.

Proof. If $c_\mu \neq 0$, give every vertex the list $\{1, 2, 3\}$ and take $L'(u) = L'(v) = \{1\}$. 3.3 gives a proper colouring with $c(u) = c(v) = 1$, impossible for adjacent vertices. □

Lemma 3.5 (counting the spare room)

Let $m, n \geq 3$ and $\ell \geq 2$.
(a) The torus $C_m \square C_n$ has spare room 0.
(b) The strip $C_m \square P_\ell$ (columns are cycles, rows are paths) has spare room m.
(c) $C_5 \square C_n$ ($n \geq 5$) with a 3×2 block of vertices removed has spare room 5.

Proof. (a) $N = mn$ and every vertex has degree 4, so $|E| = 2N$.

(b) $N = m\ell$. The ℓ columns are copies of C_m: $m\ell$ edges. Each of the $\ell - 1$ pairs of consecutive columns is joined by m row edges: $m(\ell - 1)$ edges. Total $2m\ell - m = 2N - m$.

(c) The block's 6 vertices have degree sum 24. The block contains 7 edges (2 vertical in each of its 2 columns, 3 horizontal), so $24 - 14 = 10$ edges leave it. Removing the block deletes 17 edges, leaving $5n - 6$ vertices and $10n - 17 = 2(5n - 6) - 5$ edges. □

So the odd torus has no room at all, and 1.4 says its only candidate coefficient is 0. A strip of a torus with m rows has exactly m units, which we spend on its two end columns, next to the deleted part.

3.2 The strip lemma for triangles

Lemma 3.6 (odd strip lemma, $m = 3$)

Let $\ell \geq 3$ be odd. In $C_3 \square P_\ell$ put weight 2 on one vertex of one end column and weight 1 on a vertex in a *different row* of the other end column. Then $c_\mu = \pm 3^{(\ell-1)/2} \neq 0$.

Proof. **Set-up.** Write the vertices as (i, c), with row $i \in \mathbb{Z}_3$ and column $c \in \{1, \ldots, \ell\}$, so $N = 3\ell$. Orient each column edge from (i, c) to $(i+1, c)$ and each row edge from (i, c) to $(i, c+1)$. (Another orientation multiplies c_μ by ± 1, which does not affect whether it is zero.) Then

$$P = \prod_{c=1}^{\ell} \prod_{i \in \mathbb{Z}_3} (x_{i,c} - x_{i+1,c}) \cdot \prod_{c=1}^{\ell-1} \prod_{i \in \mathbb{Z}_3} (x_{i,c} - x_{i,c+1}).$$

Let ω be a primitive cube root of unity, so $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. Exponents of ω are read mod 3. Let the shape put weight 2 at row r of one end column and weight 1 at row $r' \neq r$ of the other, and let $k \in \{1, 2\}$ be the weight on column ℓ, so column 1 carries weight $3 - k$.

Part 1: from the coefficient to a sum.

Step 1. *Why multiply by* $\prod x_v^{\mu(v)}$. With sets of size 3, 2.4 can only read the coefficient of $\prod_v x_v^2$ (it needs $|A_v| = t_v + 1$, so $t_v = 2$). Put $Q = P \cdot \prod_v x_v^{\mu(v)}$. Multiplying by a fixed monomial sends distinct monomials to distinct monomials, and sends $\prod_v x_v^{2-\mu(v)}$ to $\prod_v x_v^2$. Hence

$$c_\mu = [\textstyle\prod_v x_v^{2-\mu(v)}]P = [\prod_v x_v^2]Q.$$

The degree condition of 2.4 holds: $\deg Q = |E| + \sum_v \mu(v) = (2N-3)+3 = 2N = \sum_v 2$, by 3.1 and 3.5(b).

Step 2. *Why the cube roots of unity.* 2.4 holds for *any* sets of size 3, and the coefficient does not depend on the choice, so we choose $A_v = \{1, \omega, \omega^2\}$ for every v. Then $\varphi_v(z) = (z-1)(z-\omega)(z-\omega^2) = z^3 - 1$ and $\varphi_v'(a) = 3a^2$. Since $a^3 = 1$, $a^2 = 1/a$, so

$$\frac{1}{\varphi_v'(a)} = \frac{a}{3}.$$

Step 3. *The sum.* By 2.4, with $Q(a) = P(a)\prod_v a_v^{\mu(v)}$:

$$c_\mu = \sum_a Q(a) \prod_v \frac{a_v}{3} = 3^{-3\ell} \sum_a P(a) \prod_v a_v^{1+\mu(v)}, \tag{$*$}$$

where a runs over all assignments $V \to \{1, \omega, \omega^2\}$.

Part 2: evaluating the sum. We think of a as a colouring with the three colours $1, \omega, \omega^2$.

Step 4. *Only proper colourings count.* $P(a)$ is a product over edges of $a_u - a_v$, so $P(a) = 0$ unless neighbours get different values. In particular each column triangle uses all three colours, so $i \mapsto$ (exponent of $a_{i,c}$) is a bijection $\mathbb{Z}_3 \to \mathbb{Z}_3$. The six maps $i \mapsto \alpha + \varepsilon i$ with $\alpha \in \mathbb{Z}_3$, $\varepsilon = \pm 1$ are distinct bijections, and there are only $3! = 6$ bijections, so these are all of them. Hence every column of a nonzero term has the form

$$a_{i,c} = \omega^{\alpha_c + \varepsilon_c i} \qquad (\alpha_c \in \mathbb{Z}_3,\ \varepsilon_c = \pm 1).$$

Step 5. *One column's three edges.* Using $\omega^x - \omega^y = \omega^x(1 - \omega^{y-x})$:

$$\prod_i (\omega^{\alpha+\varepsilon i} - \omega^{\alpha+\varepsilon(i+1)}) = \omega^{3\alpha+\varepsilon(0+1+2)}(1-\omega^\varepsilon)^3 = (1-\omega^\varepsilon)^3,$$

since $3\alpha \equiv 0$ and $0+1+2 = 3 \equiv 0$. Now $(1-\omega)^2 = 1 - 2\omega + \omega^2 = (1+\omega+\omega^2) - 3\omega = -3\omega$, so $(1-\omega)^3 = -3\omega(1-\omega) = -3(\omega - \omega^2)$. Put $g = 3(\omega - \omega^2)$. Then $(1-\omega)^3 = -g$, and taking complex conjugates ($\bar\omega = \omega^2$, $\bar g = -g$) gives $(1-\omega^2)^3 = g$. In one formula:

$$\text{column factor} = (1-\omega^\varepsilon)^3 = -\varepsilon g.$$

Step 6. *The three row edges between columns c and $c+1$.* Write $\alpha = \alpha_c$, $\beta = \alpha_{c+1}$.

If $\varepsilon_{c+1} = -\varepsilon_c$, rows i of the two columns have equal colours when $\alpha + \varepsilon i \equiv \beta - \varepsilon i$, that is $2\varepsilon i \equiv \beta - \alpha$. Since 2 and ε are invertible mod 3, this has a solution i, so one factor is 0. Hence in every nonzero term *all columns have the same* ε.

If $\varepsilon_{c+1} = \varepsilon_c = \varepsilon$, put $\delta = \beta - \alpha$. Then

$$\prod_i (\omega^{\alpha+\varepsilon i} - \omega^{\beta+\varepsilon i}) = \omega^{3\alpha+3\varepsilon}(1-\omega^\delta)^3 = (1-\omega^\delta)^3 = \begin{cases} 0 & \delta = 0, \\ -g & \delta = 1, \\ g & \delta = 2, \end{cases}$$

by the previous step. We write this as $g\,\chi(\delta)$ with $\chi(0) = 0$, $\chi(1) = -1$, $\chi(2) = 1$. It does not depend on ε.

Step 7. *The vertex weights.* Split $\prod_v a_v^{1+\mu(v)} = \prod_v a_v \cdot \prod_v a_v^{\mu(v)}$. Each column contains $1, \omega, \omega^2$ once, with product $\omega^3 = 1$, so $\prod_v a_v = 1$. Only the two weighted vertices remain:

$$\prod_v a_v^{\mu(v)} = \omega^{\sum_v \mu(v)(\alpha_{\mathrm{col}(v)} + \varepsilon\,\mathrm{row}(v))} = \omega^{(3-k)\alpha_1 + k\alpha_\ell + \varepsilon s} = \omega^{k(\alpha_\ell - \alpha_1) + \varepsilon s},$$

where $s = \sum_v \mu(v)\,\mathrm{row}(v) \bmod 3$, and $(3-k)\alpha_1 \equiv -k\alpha_1$.

Step 8. *Summing.* By Steps 4 to 7, a nonzero term is fixed by ε and $\alpha_1, \dots, \alpha_\ell$, and equals

$$\underbrace{(-\varepsilon g)^\ell}_{\text{columns}} \cdot \underbrace{\prod_{c=1}^{\ell-1} g\,\chi(\delta_c)}_{\text{rows}} \cdot \omega^{k(\delta_1 + \cdots + \delta_{\ell-1}) + \varepsilon s}, \qquad \delta_c = \alpha_{c+1} - \alpha_c,$$

using $\alpha_\ell - \alpha_1 = \delta_1 + \cdots + \delta_{\ell-1}$. The map $(\alpha_1, \dots, \alpha_\ell) \mapsto (\alpha_1, \delta_1, \dots, \delta_{\ell-1})$ is a bijection of $\mathbb{Z}_3^\ell$, and the term does not involve α_1 (3 choices). So the sum factorises:

$$c_\mu = 3^{-3\ell}\, g^{2\ell-1} \cdot \Big(\sum_{\varepsilon = \pm 1} (-\varepsilon)^\ell \omega^{\varepsilon s} \Big) \cdot 3 \cdot \Big(\sum_{\delta \in \mathbb{Z}_3} \chi(\delta)\, \omega^{k\delta} \Big)^{\ell-1}.$$

Since ℓ is odd, $(-\varepsilon)^\ell = -\varepsilon$, so the first bracket is $-\omega^s + \omega^{-s} = \omega^{2s} - \omega^s$. The second bracket is $0 - \omega^k + \omega^{2k} = \omega^{2k} - \omega^k$. Therefore

$$c_\mu = 3^{1-3\ell}\, g^{2\ell-1}\, (\omega^{2s} - \omega^s)\, (\omega^{2k} - \omega^k)^{\ell-1}.$$

Step 9. *Nonzero, and the value.* $g \neq 0$, and $\omega^{2t} - \omega^t = \omega^t(\omega^t - 1)$ is 0 exactly when $t \equiv 0$. Here $k \in \{1, 2\}$, and $s = 2r + r' \equiv r' - r \not\equiv 0$ because the rows differ. So $c_\mu \neq 0$. For the value, $|\omega - \omega^2| = \sqrt{3}$, so $|g| = 3\sqrt{3}$ and both brackets have modulus $\sqrt{3}$:

$$|c_\mu| = 3^{1-3\ell}\, (3\sqrt{3})^{2\ell-1}\, \sqrt{3}\, (\sqrt{3})^{\ell-1} = 3^{\,1-3\ell+\frac{3}{2}(2\ell-1)+\frac{1}{2}+\frac{\ell-1}{2}} = 3^{(\ell-1)/2}.$$

c_μ is a coefficient of a polynomial with integer coefficients, so it is an integer: $c_\mu = \pm 3^{(\ell-1)/2}$. □

Note. Every formula above is checked by exact arithmetic. The script is `strip3_closed_form.py` in `odd-torus/ws-W7`. It checks the column and row factors for all $\alpha, \beta, \varepsilon$; the sum $(*)$ against the closed form for all 12 shapes at $\ell = 3, 5, 7$; and, for $\ell = 3$, $(*)$ against the coefficient read directly from the expanded polynomial.

3.3 The prism lemma

The two deleted columns form a *prism*: triangles $u_0u_1u_2$ and $v_0v_1v_2$ joined by the edges u_tv_t.

Computer-assisted result 3.7 (two-column lemma; `ws-W9/prism_sizes.c`)

Give u_0 a list of size 3 and every other prism vertex a list of size 2 or 3. If the prism has no colouring, then $|B_1| = |B_2| = 2$ and $B_1 = B_2$, where B_t is the list of v_t.

Note. Proof by checking every list assignment up to renaming colours (236,160,170 cases, 8,335 not colourable, all with $B_1 = B_2$). Amrutanshu has a hand proof of the sharper statement that failure happens exactly in two patterns, (Z) and (W); this was also confirmed exhaustively. Merging the hand proof is a to-do.

3.4 The theorem

Theorem 3.8 (Theorem A)

$C_3 \,\square\, C_n$ is 3-choosable for every $n \geq 3$. Hence $\chi_\ell(C_3 \,\square\, C_n) = 3$.

Proof. Even n: 1.4. $n = 3$: 1.5. Let $n \geq 5$ be odd and $|L(v)| = 3$ for all v.

Case 1: some row list changes. There are r, j and $\alpha \in L(r, j) \setminus L(r, j+1)$. Name columns $C_0 = j, C_1, C_2, C_3$; let r', r'' be the other two rows.

Step 1. Choose β: if $|L(r', C_2) \cap L(r'', C_2)| = 2$, let β be the colour of $L(r', C_2) \setminus L(r'', C_2)$; otherwise any colour.

Step 2. Delete C_1, C_2. The rest is a strip of odd length $n - 2 \geq 3$. Pin (r, C_0) to α and forbid β at (r', C_3). By 3.6 the strip has such a colouring ψ.

Step 3. The prism $C_1 \cup C_2$ gets lists $A_t = L(t, C_1) \setminus \{\psi(t, C_0)\}$ and $B_t = L(t, C_2) \setminus \{\psi(t, C_3)\}$. A_r keeps 3 colours since $\alpha \notin L(r, C_1)$.

Step 4. If the prism fails, 3.7 gives $B_{r'} = B_{r''} = E$, a 2-set. Then $L(r', C_2) = E \cup \{\psi(r', C_3)\}$ and $L(r'', C_2) = E \cup \{\psi(r'', C_3)\}$, with the two extra colours different (their vertices are adjacent in C_3). So $\psi(r', C_3)$ is the colour of $L(r', C_2) \setminus L(r'', C_2)$, which is β. But β was forbidden there. Contradiction.

Case 2: no row list changes. Then every row r has a constant list A_r. By 1.5, colour $C_3 \,\square\, C_3$ from these row lists; call its columns X, Y, Z. They are pairwise compatible (different in every row). Colour the columns of $C_3 \,\square\, C_n$ as $X, Y, Z, Y, Z, \ldots, Y, Z$: since n is odd the last column is Z, which is compatible with X. □

Known result 3.9 (Amrutanshu, kernel method; checked by us)

For odd k, $C_3 \,\square\, C_{3k}$ has a kernel-perfect orientation with all out-degrees 2. Hence it is 3-choosable and even 3-paintable.

Think it over

Q1. Where exactly does the proof of 3.8 use that n is odd? Where does it use $n \geq 5$?

Q2. Why can the polynomial method not prove 3.8 on its own, by 1.4?

Q3. In Case 1, why is it enough to forbid a *single* colour β?

Chapter 4

General lemmas for every odd m

From now on m, n are odd and $m \geq 5$. These lemmas are proved by hand and hold for every palette.

4.1 Special shapes of lists

Corollary 4.1 (constant rows or columns)

If every row has a constant list, or every column has a constant list, $C_m \square C_n$ is L-colourable.

Proof. Rows constant, $L(i,j) = A_i$: $C_m \,\square\, C_3 \cong C_3 \,\square\, C_m$ is 3-choosable by 3.8; colour it from the row lists to get pairwise compatible columns X, Y, Z, and use $X, Y, Z, Y, Z, \ldots$ as in Case 2. Columns constant: the same with rows and columns exchanged. □

Lemma 4.2 (one free row)

If all rows except one have the same constant list A, $C_m \,\square\, C_n$ is L-colourable.

Proof. Let the free row be i_0. The other rows form a path of $m-1$ rows (an even number, at least 4).

Step 1. Choose a proper colouring p of this path with values in $\mathbb{Z}_3$ whose two ends are equal: $0, 1, 2, 0$ followed by $1, 0, 1, 0, \ldots$.

Step 2. Choose $\sigma_j \in \{+1, -1\}$ with $\sum_j \sigma_j \equiv 0 \pmod 3$ (take k values $+1$ with $2k \equiv n$, $k \in \{0, 1, 2\}$). Let $t_0 = 0$, $t_{j+1} = t_j + \sigma_j$, fix a bijection $\pi : \mathbb{Z}_3 \to A$, and colour (i, j) with $\pi(p(i) + t_j)$.

Step 3. Columns are proper because p is; rows are proper because $\sigma_j \neq 0$. Both neighbours of (i_0, j) get the same colour a_j, so row i_0 has lists $L(i_0, j) \setminus \{a_j\}$ of size at least 2.

Step 4. By 2.2(a) row i_0 can be coloured unless all these lists are one 2-set B. Then $L(i_0, j) = B \cup \{a_j\}$, and the a_j take at least two values, all outside B. Replace π by $\tau \circ \pi$ for a 3-cycle τ of A: if row i_0 still failed, every $\tau(a_j)$ would lie in $B \cap A$, forcing $|A \setminus B| \leq 1$, which contradicts the two values. □

4.2 Merging colours

Lemma 4.3 (merging, Lemma M)

Suppose colours c and d never appear together in one list. Replace d by c in every list, giving L' (still lists of size 3). If G is L'-colourable, then G is L-colourable.

Proof. Take an L'-colouring. A vertex coloured c had exactly one of c, d in $L(v)$; give it that one. If two neighbours now had the same colour, they had the same colour in L' too, which is impossible. □

Corollary 4.4

If $C_m \,\square\, C_n$ is not 3-choosable, it has a bad list assignment in which every pair of colours appears together in some list. Such an assignment uses at most k colours, where $\binom{k}{2} \le 3mn$. For $C_5 \,\square\, C_5$ this gives at most 12 colours.

4.3 Projection and the sea

Definition 4.5 (colour graph). The *colour graph* $K(L)$ has the colours as vertices; two colours are joined if they lie in a common list. Every list is a triangle of $K(L)$.

Lemma 4.6 (projection)

If $K(L)$ is 3-colourable, $C_m \,\square\, C_n$ is L-colourable.

Proof. Let π be a proper 3-colouring of $K(L)$ with values in $\mathbb{Z}_3$; it is one-to-one on every list. Let ψ be a proper 3-colouring of the torus with values in $\mathbb{Z}_3$ (1.6). Give v the unique $c \in L(v)$ with $\pi(c) = \psi(v)$. Neighbours have different ψ, hence different π-values, hence different colours. □

Lemma 4.7 (sea lemma)

Let X be a set of 3 colours. Call v a *defect* if $L(v) \neq X$, and let $O(v) = L(v) \setminus X$ (never empty for a defect). Suppose there is a set T of defects such that

(a) the defects not in T can be properly coloured using only their colours $O(v)$;
(b) T has at most one vertex in each column;
(c) every $v \in T$ has exactly one colour outside X, so $L(v) \cap X$ misses exactly one colour ϕ_v of X;
(d) it is not the case that T meets every column, lies in a single row, and has all ϕ_v equal.

Then $C_m \,\square\, C_n$ is L-colourable. (The same holds with "column" and "row" exchanged in (b) and (d).)

Proof. Identify X with $\mathbb{Z}_3$. Use a *sum colouring* $\psi(i,j) = f(i) + g(j)$, where f is a proper colouring of C_m and g a proper colouring of C_n; it is a proper 3-colouring of the torus.

Step 1. Vertices with list X get $\psi(v)$. Defects not in T get their colour from (a): they never clash with X-coloured vertices, and not with each other.

Step 2. A vertex $v = (i, j) \in T$ gets $\psi(v)$, which needs $\psi(v) \neq \phi_v$, that is, $g(j) \neq \phi_v - f(i)$. By (b) each column loses at most one value for g.

Step 3. So g must colour the odd cycle C_n from lists of size 2 or 3. By 2.2 this works unless every column has a T-vertex and the values $\phi_v - f(i_v)$ are all equal.

Step 4. If T uses two rows $i_1 \neq i_2$, change f: proper colourings of C_m realise both nonzero values of $f(i_1) - f(i_2)$ (all three if the rows are not adjacent), so the values can be made unequal. If T uses a single row, equality means all ϕ_v are equal, which (d) excludes. □

Note. 4.6 and 4.7 combine: replace "list equal to X" by "list mapped one-to-one onto $\mathbb{Z}_3$ by a fixed π". The proofs are the same.

Think it over

Q1. In 4.3, what goes wrong if c and d share a list?

Q2. Explain how the idea of 4.6 contains the case "all lists equal".

Q3. Find an assignment where 4.7 applies but 4.6 does not, and one where the reverse holds.

Q4. Why is hypothesis (b) of 4.7 needed?

Chapter 5

The case $m = 5$: what is proved

5.1 Results by computation

Computer-assisted result 5.1 (edge deletion; `ws-W4`)

For $m \in \{5, 7\}$ and every odd n: $C_m \,\square\, C_n$ minus any edge is 3-choosable. Consequently, in a bad list assignment of $C_5 \,\square\, C_n$, every edge has overlapping lists, and every colour of every list also appears in a neighbour's list.

Computer-assisted result 5.2 (strip lemma, $m = 5$; `ws-W7/prove_strip5.py`)

For 74 shapes (a pin on one row of one end, three single units elsewhere), the strip coefficient of $C_5 \,\square\, P_\ell$ is nonzero for every odd $\ell \geq 3$.

Note. Method: the coefficients for odd ℓ satisfy an exact linear recurrence of order 6 (from the minimal polynomial of a 30×30 transfer matrix). For each shape the largest root has a nonzero weight and dominates from $\ell = 3$ on, with certified error bounds.

Computer-assisted result 5.3 (Theorem B; `ws-W8/alln_q4.py`)

For every odd n, $C_5 \,\square\, C_n$ is L-colourable whenever all lists are drawn from a palette of 4 colours.

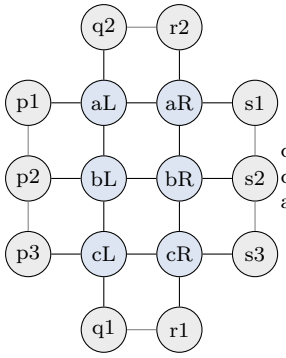
Note. Outline: every 4-colour two-column window was judged exhaustively; "every cut fails" becomes a closed walk in a finite graph (262,144 states); every part of that graph carrying an odd closed walk consists of a special shape handled by 4.1 or 4.2.

Computer-assisted result 5.4 (no small counterexample; `ws-W2/certify.py`)

$C_5 \,\square\, C_5$ has no bad list assignment from a palette of 4 colours (certified SAT result).

5.2 The hole cut

Instead of deleting whole columns, delete a 3×2 *block*: rows $t-1, t, t+1$ and columns $j, j+1$. Name the block vertices and the 10 *border* vertices as in the picture.



q1, r1: row $t+2$; q2, r2: row $t+3 = t-2$. q1 and q2 are adjacent (the column wraps around), and so are r1 and r2.

By 3.5(c) the rest of the torus has exactly 5 units of spare room. Since $n \geq 5$, the 10 edges leaving the block go to 10 distinct border vertices, each with exactly one neighbour in the block; the 5 units are spent on the border, and by 3.4 no two adjacent border vertices are both pinned.

Computer-assisted result 5.5 (hole coefficients; `ws-W14/hole_coeff.py`, `hole_prove.py`)

For 1,256 of the 1,388 ways of placing 5 units on the border (no two adjacent pins), the coefficient for $C_5 \,\square\, C_n$ minus the block is nonzero for every odd $n \geq 5$.

Note. Checked two ways: on the full torus the same code gives $\mathrm{tr}(T^n) = 0$ for odd n and a nonzero value for even n, as 1.4 predicts; and 1,800 random list assignments matching nonzero coefficients were all colourable.

Lemma 5.6 (block-fill criterion)

After the border is coloured, suppose block vertex u has lost ℓ_u of its colours. The block can be filled if, for every set S of block vertices,

$$\sum_{u \in S} \ell_u \;\leq\; 2|S| - e(S),$$

where $e(S)$ is the number of block edges with both ends in S.

Proof. The block is bipartite. By 2.6 the condition gives an orientation with $\mathrm{outdeg}(u) \leq 2 - \ell_u = |L'(u)| - 1$. By the Note after 2.5, this orientation satisfies 2.5. □

Lemma 5.7 (merging inside a hole)

Let the adversary choose border colours without respecting the edges between border vertices. If two colours never share one of the 16 lists of a hole, merging them can only turn a working strategy into a working strategy for the original. Hence a failing hole stays failing after merging, and every failing hole comes from one using at most 10 colours.

Proof. As in 4.3: strategies, adversary moves and fills transfer through the merge, and a forbidden colour cannot be dodged by its partner because the two never share a list. For the bound, $\binom{k}{2} \leq 16 \cdot 3$. □

Computer-assisted result 5.8 (isolated block; `ws-W14/block_types_allowed.c`)

If a 3×2 block has all six lists equal to X and all ten border lists different from X, then $C_5 \,\square\, C_n$ is L-colourable (every odd $n \geq 5$, every palette).

Note. Only "which colours of X a border list contains" matters for the fill, so the check is independent of the palette: all 7^{10} border types (11,790,389 up to symmetry) have a strategy using only placements from 5.5.

5.3 Seams

Definition 5.9 (piecewise row-constant). An assignment is *piecewise row-constant* if the columns split into consecutive blocks, each of at least 3 columns, such that inside block i row r has a constant list A_r^i. A *seam* is the boundary between two blocks.

Lemma 5.10 (seam lemma, light overlap)

Let L be piecewise row-constant and suppose that at every seam between blocks i and $i+1$, $\sum_r |A_r^i \cap A_r^{i+1}| \leq 8$. Then $C_5 \,\square\, C_n$ is L-colourable.

Proof. **Step 1.** For each block, colour $C_5 \,\square\, C_3$ from its row lists (3.8). Its columns X^i, Y^i, Z^i are pairwise compatible, and row r uses each colour of A_r^i exactly once.

Step 2. At a seam, pair a column P of block i with a column Q of block $i+1$ (9 pairs). A common colour $c \in A_r^i \cap A_r^{i+1}$ is at row r of exactly one P and one Q, so it spoils exactly one pair. At most 8 pairs are spoiled, so some pair (P, Q) is compatible.

Step 3. Inside a block of $\ell \geq 3$ columns, use a sequence of its three columns with neighbours different: a walk of $\ell - 1 \geq 2$ steps on a triangle, which can start and end at any prescribed corners.

Step 4. Each seam uses its compatible pair and each block joins its two ends, so every edge is proper. □

Think it over

Q1. Count again: why does deleting the block remove exactly 17 edges?

Q2. In 5.6, take S to be the whole block. What does the condition say?

Q3. Where does the proof of 5.10 use that blocks have at least 3 columns?

Q4. In 5.8, why does a border list different from X always contain a colour outside X?

Chapter 6

The case $m = 5$: the remaining gap

6.1 The reduction

Call a hole *failing* if no strategy from 5.5 works against every adversary. If some hole does not fail, the torus is coloured. So it is enough to understand assignments in which *every* hole fails.

Definition 6.1 (local sea, local seam). A hole is a *local sea* if its 16 lists satisfy the hypotheses of 4.7 (with the Note), read inside the hole. It is a *local seam* if, inside the hole, every row is constant on each side of one column boundary.

Verified, not proved 6.2 (`ws-W14/i2_test.py`)

Every failing hole is a local sea or a local seam. (646 failing holes tested: 119 sea only, 98 seam only, 429 both, 0 neither.)

Open 6.3 (gluing)

If every hole of $C_5 \,\square\, C_n$ is a local sea or a local seam, then the whole assignment satisfies 4.7 (with the Note), or is piecewise row-constant, or is one of the shapes of 4.1.

Open 6.4 (seam lemma, heavy overlap)

5.10 without the overlap condition.

Verified, not proved 6.5 (`ws-W16/seam_test.py`)

For two blocks with heavy overlap (390 tests with overlap at least 9), suitable column triples with a compatible pair always exist.

3-choosability of $C_5 \,\square\, C_n$ (odd n) would follow from: a proof of 6.2 for every palette (by 5.7 it is enough to treat holes with at most 10 colours), plus 6.3, plus 6.4. Everything else used is proved above.

6.2 Plan for each gap

- **6.4.** Given a column P of block i, a compatible column Q of block $i+1$ exists by 2.2. Q extends to a triple when the flip constraints between neighbouring rows (rows sharing a leftover colour must swap it) are consistent around the 5-cycle; they are whenever two

neighbouring rows share no leftover colour. Remaining: the fully constrained case, and closing up around all blocks.

- **6.3.** Neighbouring holes share most vertices and failing holes have at most 4 defects (in 502 of 509 tests), so their sets X agree; every column lies inside one hole, so the "one T-vertex per column" rule is checked locally. Remaining: long chains of defects, and regions where seas meet seams.
- **6.2.** By 5.7 only holes with at most 10 colours matter, and 5.6 turns the fill into counting losses. Remaining: a case analysis by how much each border list shares with its block neighbour's list (one colour: one forbid neutralises it; two colours: pin it to its third colour; all three: it cannot be neutralised).

Think it over

Q1. Explain why "every hole fails" is the only case left, given 5.5.

Q2. Why does 5.7 need the adversary to ignore the edges between border vertices?

Q3. Give an example of a failing hole that is a seam but not a sea. (See Chapter 6.)

Chapter 7

What did not work, and why

These are dead ends we checked; they explain the shape of the reduction.

1. **Two-column cuts with one shape.** Amrutanshu's shape (pin, then forbid at the far side outside two rows p, q with $2r \not\equiv p+q$) fails on 6,904 of the 4^{10} four-colour windows. All 74 shapes, cuts in both directions and 4.2 were needed for 5.3.

2. **Bad windows are not "mostly constant" in a simple count.** The guess "at least 7 of 10 lists are equal" is false (3 counterexamples). Every bad window found has at least 3 rows whose two lists are equal.

3. **No short list of prism failures for 5-cycles.** Unlike 3.7, failures of the $C_5 \,\square\, K_2$ fill come in many shapes.

4. **Three-column cuts and double pins.** Better on generic lists, but they also fail on nearly constant lists.

5. **Nearly constant lists beat every local cut.** This is why 4.7, 5.10 and 5.8 exist: such lists must be coloured by a global pattern.

6. **A failing hole need not be a sea.** The seam below fails as a hole (each row read left to right):

row	column $j-1$	j	$j+1$	$j+2$
$t+3$		013	014	
$t-1$	013	013	014	014
t	123	123	012	012
$t+1$	134	134	013	013
$t+2$		134	013	

Think it over

Q1. In the table, check that every row is constant on each side of the boundary between j and $j+1$.

Q2. Why does a hole placed exactly on a seam give the adversary full control of the border?

Chapter 8

Summary and plan

m or n even	1.4 (known)
$C_3 \,\square\, C_n$	3.8 (proved; one finite lemma by exhaustive check)
$C_5 \,\square\, C_n$, palette 4	5.3 (computer-assisted)
$C_5 \,\square\, C_n$, any palette	reduced to 6.2, 6.3 and 6.4
all odd m	lemmas of Chapter 4 hold for all odd m; strip and hole coefficients only for $m = 5$

Order of work.

1. This week: submit 3.8 (merge the hand proof of 3.7; literature check).
2. Next: 6.4 (closest to done), then 6.3, then 6.2.
3. After $m = 5$: a strip lemma uniform in m; decide by mid-December whether to do $m = 7$ by computer.

Chapter A

Where every computation lives

All paths are inside `~/work/gt/odd-torus/`.

Result	Script	Output
3.6 (closed form check)	`ws-W1/strip_coeff.py`, `compare.py`	0 mismatches in 500
3.7	`ws-W9/prism_sizes.c`, `prism.c`	0 exceptions
4.2 (construction)	`ws-W9/freerow_check.py`	0 failures in 20,000
5.1	`ws-W4/`	see `REPORT.md`
5.2	`ws-W7/prove_strip5.py`	148 cases
5.3	`ws-W8/alln_q4.py`, `validate_shapes.py`	81 components; 40 of 40 agree
5.4	`ws-W2/certify.py`	UNSAT, certified
5.5	`ws-W14/hole_coeff.py`, `hole_prove.py`, `hole_check.py`	1,256 placements
5.8	`ws-W14/block_types_allowed.c`	0 without strategy
4.7 (construction)	`ws-W15/sea_check.py`	3,810 of 3,810
6.2	`ws-W14/i2_test.py`	0 neither
6.5	`ws-W16/seam_test.py`	0 failures in 390

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